

Task 1 — Document 08: Module Definition Specification

Problem Statement ID: PS 26117 / SIH26117 **Title:** Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work **Organization:** Mangalore Refinery and Petrochemicals Limited (MRPL)
Category: Software | **Theme:** Smart Automation

■ Software Module Architecture

The **Sovereign AI Workbench** codebase is structured into six decoupled, highly cohesive software modules. Each module maintains clean API boundaries to allow independent maintenance and future extensibility.

```
sovereign-ai/
├── frontend/
│   ├── index.html # [Module 1] Web Interface Module
│   ├── main.py # [Module 2 & 6] FastAPI Backend Gateway & Guardrails
│   ├── kb_index.py # [Module 3 & 4] Document/OCR Ingestion & Vector Indexer
│   ├── kb_search.py # [Module 4] Cosine Vector Search Engine
│   ├── rag_test.py # [Module 5 & 6] End-to-End RAG Verification Module
│   └── data/ # [Module 4] Storage Module
│       ├── knowledge_base/ # Source Files & Images
│       └── kb_index.json # Vector Index Storage
```

■ Module Specifications

Module 1: Web Interface Module (`frontend/index.html`)

- **Responsibility:** Provides the responsive, dark-themed Single Page Application (SPA) user interface.
- **Key Components:**
 - **Chat Window:** Renders user messages and formatted markdown AI responses.
 - **Document Uploader:** Form payload handler for drag-and-drop file uploads.
 - **Knowledge Toggle:** Interactive UI switch modifying payload state (`knowledge_enabled: true/false`).
 - **Source Badges:** Renders document pill badges showing source file names and cosine similarity scores.
- **Input:** User clicks, file selections, keyboard inputs.
- **Output:** REST HTTP calls to FastAPI backend; DOM updates.

Module 2: FastAPI Gateway & Task Router Module (`main.py`)

- **Responsibility:** Serves as the central API gateway, managing request endpoints and dynamic prompt routing.
- **Endpoints:**
 - **GET /:** Serves the static single-page Web UI.
 - **POST /upload:** Ingests document/image upload streams, routes to parser, appends vectors to KB index.
 - **POST /chat:** Receives chat messages, dispatches to Model Router or RAG Engine, returns JSON answers.
- **Router Logic:**
 - Coding keyword match → `route: coding` → `qwen2.5-coder:7b`.
 - `knowledge_enabled == true` → `route: rag` → Vector Search → Grounded `qwen3.5:4b`.
 - Default → `route: general` → Direct `qwen3.5:4b`.

Module 3: Multimodal Document & Image Parsing Engine

- **Responsibility:** Handles file decoding and text extraction across heterogeneous file types.
 - **Supported File Adapters:**
 - **PDF Adapter:** Uses `pypdf.PdfReader` to extract text from multi-page PDFs.
 - **Text/Markdown Adapter:** Native UTF-8 string file decoding.
 - **OCR Adapter:** Encodes `.png`, `.jpg`, `.jpeg`, `.webp` into Ollama chat payload, calls `glm-ocr:q8_0` for visual text extraction.
 - **Input:** File streams saved to `data/knowledge_base/`.
 - **Output:** Cleaned normalized text string.
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Module 4: Vector Indexing & Cosine Search Engine (`kb_index.py` & `kb_search.py`)

- **Responsibility:** Generates text embeddings, maintains vector store state, and executes high-speed similarity search.
 - **Core Operations:**
 - **Chunking:** Splits normalized text into 500-character windows with 50-character overlap.
 - **Embedding Generation:** Calls local `nomic-embed-text` via Ollama Python client to generate 768-dimensional float arrays.
 - **Index Persistence:** Writes JSON structure to `data/kb_index.json`.
 - **Cosine Search:** Performs dot-product cosine similarity vector math between query vector and index vectors to return Top-K matching snippets.
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Module 5: Local LLM Runtime Manager (Ollama Interface)

- **Responsibility:** Manages IPC communications with local Ollama daemon.
 - **Hosted Models:**
 - `qwen3.5:4b`: General reasoning & RAG synthesis.
 - `qwen2.5-coder:7b`: Code generation.
 - `glm-ocr:q8_0`: OCR text extraction.
 - `nomic-embed-text`: Text embedding generation.
 - **Settings:** Thinking mode turned OFF (`--think=false`) for low latency inference.
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Module 6: Grounding & Anti-Hallucination Guardrail Module

- **Responsibility:** Enforces truthfulness and strict context boundaries during RAG generation.
- **Guardrail Rule:**
- System prompt enforces: *"Answer ONLY based on the provided context. If the answer cannot be found in the context, explicitly state 'That information is not available in the knowledge base.'"*
- Rejects speculative answers when retrieval score is below threshold.